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Topology-based retirement in a memory system

Abstract

Methods, systems, and devices for topology-based retirement in a memory system are described. In some examples, a memory system or memory device may be configured to evaluate error conditions relative to a physical or electrical organization of a memory array, which may support inferring the presence or absence of defects in one or more structures of a memory device. For example, based on various evaluations of detected errors, a memory system or a memory device may be able to infer a presence of a short-circuit, an open circuit, a dielectric breakdown, or other defects of a memory array that may be related to wear or degradation over time, and retire a portion of a memory array based on such an inference.

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Background/Summary

CROSS REFERENCE (1) The present Application for Patent is a continuation of U.S. patent application Ser. No. 17/574,024 by Yeung et al., entitled “TOPOLOGY-BASED RETIREMENT IN A MEMORY SYSTEM”, filed Jan. 12, 2022, which claims priority to U.S. Provisional Patent Application No. 63/147,679 by Yeung et al., entitled “TOPOLOGY-BASED RETIREMENT IN A MEMORY SYSTEM”, filed Feb. 9, 2021, each of which is assigned to the assignee hereof and is expressly incorporated by reference in its entirety herein.

FIELD OF TECHNOLOGY
(1) The following relates generally to one or more systems for memory and more specifically to topology-based retirement in a memory system.
BACKGROUND
(2) Memory devices are widely used to store information in various electronic devices such as computers, wireless communication devices, cameras, digital displays, and the like. Information is stored by programing memory cells within a memory device to various states. For example, binary memory cells may be programmed to one of two supported states, often corresponding to a logic 1 or a logic 0. In some examples, a single memory cell may support more than two possible states, any one of which may be stored by the memory cell. To access information stored by a memory device, a component may read, or sense, the state of one or more memory cells within the memory device. To store information, a component may write, or program, one or more memory cells within the memory device to corresponding states.
(3) Various types of memory devices exist, including magnetic hard disks, random access memory (RAM), read-only memory (ROM), dynamic RAM (DRAM), synchronous dynamic RAM (SDRAM), static RAM (SRAM), ferroelectric RAM (FeRAM), magnetic RAM (MRAM), resistive RAM (RRAM), flash memory, phase change memory (PCM), 3-dimensional cross-point memory (3D cross point), not-or (NOR) and not-and (NAND) memory devices, and others. Memory devices may be volatile or non-volatile. Volatile memory cells (e.g., DRAM cells) may lose their programmed states over time unless they are periodically refreshed by an external power source. Non-volatile memory cells (e.g., NAND memory cells) may maintain their programmed states for extended periods of time even in the absence of an external power source.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** illustrates an example of a system that supports topology-based retirement in a memory system in accordance with examples as disclosed herein.
(2) FIG. **2** shows a block diagram of a circuit that supports topology-based retirement in a memory system in accordance with examples as disclosed herein.
(3) FIG. **3** shows a process flow illustrating a method that supports topology-based retirement in a memory system in accordance with examples as disclosed herein.
(4) FIG. **4** shows a block diagram of a memory system that supports topology-based retirement in a

memory system in accordance with examples as disclosed herein.
(5) FIG. **5** shows a flowchart illustrating a method or methods that support topology-based retirement in a memory system in accordance with examples as disclosed herein.
DETAILED DESCRIPTION
(6) Memory systems may include one or more memory devices that include an array of memory cells and circuitry operable to perform access operations on the memory cells. Various structures of a memory device (e.g., of a memory die) may wear or degrade over time, which may lead to operational failures or otherwise unreliable operation of the memory device. Some memory systems or memory devices may be configured to detect indications of unreliable or failed access operations, and retire portions of a memory array in response to such detections. However, some techniques for detecting unreliable or failed access operations may be overly conservative and lead to excessive retirement of a memory array.
(7) In accordance with examples as disclosed herein, a memory system or memory device may be configured to perform topology-based evaluations for decisions on whether to retire portions of a memory array. For example, a memory system or memory device may be configured to evaluate error conditions relative to a physical or electrical organization of a memory array, which may support inferring the presence or absence of defects (e.g., physical defects, material defects, electrical defects) in one or more structures of a memory device. In some cases, the errors observed in a memory array may be permanent errors that warrant retiring a block or the errors observed in a memory array may be more transient errors that may be resolved. To avoid retiring blocks of memory due to errors that are more transient in nature, techniques may use multiple characterizations of the error to determine whether to retire a block. For example, based on various evaluations of detected errors, a memory system or a memory device may be able to infer a presence of a short-circuit, an open circuit, a dielectric breakdown, or other defects of a memory array that may be related to wear or degradation over time and, in response, may retire a portion of a memory array based on such an inference. Compared to other techniques of array retirement, by implementing one or more aspects of topology-based retirement in accordance with examples as disclosed herein, a memory device may be configured with a larger capacity, a smaller degree of over-provisioning, or a longer life cycle, among other benefits or combinations thereof.
(8) Features of the disclosure are initially described in the context of systems, devices, and circuits with reference to FIGS. **1** and **2**. Features of the disclosure are described in the context of a flowchart for topology-based retirement with reference to FIG. **3**. These and other features of the disclosure are further illustrated by and described with reference to an apparatus diagram and a flowchart that relate to topology-based retirement in a memory system as described with reference to FIGS. **4** and **5**.
(9) FIG. **1** illustrates an example of a system **100** that supports topology-based retirement in a memory system in accordance with examples as disclosed herein. The system **100** includes a host system **105** coupled with a memory system **110**.
(10) A memory system **110** may be or include any device or collection of devices, where the device or collection of devices includes at least one memory array. For example, a memory system **110** may be or include a Universal Flash Storage (UFS) device, an embedded Multi-Media Controller (eMMC) device, a flash device, a universal serial bus (USB) flash device, a secure digital (SD) card, a solid-state drive (SSD), a hard disk drive (HDD), a dual in-line memory module (DIMM), a small outline DIMM (SO-DIMM), or a non-volatile DIMM (NVDIMM), among other possibilities.
(11) The system **100** may be included in a computing device such as a desktop computer, a laptop computer, a network server, a mobile device, a vehicle (e.g., airplane, drone, train, automobile, or other conveyance), an Internet of Things (IoT) enabled device, an embedded computer (e.g., one included in a vehicle, industrial equipment, or a networked commercial device), or any other computing device that includes memory and a processing device.
(12) In some examples, a coupling between the host system **105** and the memory system **110** may

include an interface with a host system controller **106**, which may be an example of a controller or control component configured to cause the host system **105** to perform various operations in accordance with examples as described herein. The host system **105** may include one or more devices, and in some cases may include a processor chipset and a software stack executed by the processor chipset. For example, the host system **105** may include an application configured for communicating with the memory system **110** or a device therein. The processor chipset may include one or more cores, one or more caches (e.g., memory local to or included in the host system **105**), a memory controller (e.g., NVDIMM controller), and a storage protocol controller (e.g., peripheral component interconnect express (PCIe) controller, serial advanced technology attachment (SATA) controller). The host system **105** may use the memory system **110**, for example, to write data to the memory system **110** and read data from the memory system **110**. Although one memory system **110** is shown in FIG. **1**, the host system **105** may be coupled with any quantity of memory systems **110**.

(13) The host system **105** may be coupled with the memory system **110** via at least one physical host interface. The host system **105** and the memory system **110** may in some cases be configured to communicate via a physical host interface using an associated protocol (e.g., to exchange or otherwise communicate control, address, data, and other signals between the memory system **110** and the host system **105**). Examples of a physical host interface may include, but are not limited to, a SATA interface, a UFS interface, an eMMC interface, a PCIe interface, a USB interface, a Fiber Channel interface, a Small Computer System Interface (SCSI), a Serial Attached SCSI (SAS), a Double Data Rate (DDR) interface, a DIMM interface (e.g., DIMM socket interface that supports DDR), an Open NAND Flash Interface (ONFI), and a Low Power Double Data Rate (LPDDR) interface. In some examples, one or more such interfaces may be included in or otherwise supported between a host system controller **106** of the host system **105** and a memory system controller **115** of the memory system **110**. In some examples, the host system **105** may be coupled with the memory system **110** (e.g., the host system controller **106** may be coupled with the memory system controller **115**) via a respective physical host interface for each memory device **130** included in the memory system **110**, or via a respective physical host interface for each type of memory device **130** included in the memory system **110**.

(14) The memory system **110** may include a memory system controller **115** and one or more memory devices **130**. A memory device **130** may include one or more memory arrays of any type of memory cells (e.g., non-volatile memory cells, volatile memory cells, or any combination thereof). Although two memory devices **130**-*a* and **130**-*b* are shown in the example of FIG. **1**, the memory system **110** may include any quantity of memory devices **130**. Further, if the memory system **110** includes more than one memory device **130**, different memory devices **130** within the memory system **110** may include the same or different types of memory cells.

(15) The memory system controller **115** may be coupled with and communicate with the host system **105** (e.g., via the physical host interface) and may be an example of a controller or control component configured to cause the memory system **110** to perform various operations in accordance with examples as described herein. The memory system controller **115** may also be coupled with and communicate with memory devices **130** to perform operations such as reading data, writing data, erasing data, or refreshing data at a memory device **130**—among other such operations—which may generically be referred to as access operations. In some cases, the memory system controller **115** may receive commands from the host system **105** and communicate with one or more memory devices **130** to execute such commands (e.g., at memory arrays within the one or more memory devices **130**). For example, the memory system controller **115** may receive commands or operations from the host system **105** and may convert the commands or operations into instructions or appropriate commands to achieve the desired access of the memory devices **130**. In some cases, the memory system controller **115** may exchange data with the host system **105** and with one or more memory devices **130** (e.g., in response to or otherwise in association with

commands from the host system **105**). For example, the memory system controller **115** may convert responses (e.g., data packets or other signals) associated with the memory devices **130** into corresponding signals for the host system **105**.
(16) The memory system controller **115** may be configured for other operations associated with the memory devices **130**. For example, the memory system controller **115** may execute or manage operations such as wear-leveling operations, garbage collection operations, error control operations such as error-detecting operations or error-correcting operations, encryption operations, caching operations, media management operations, background refresh, health monitoring, and address translations between logical addresses (e.g., logical block addresses) associated with commands from the host system **105** and physical addresses (e.g., physical block addresses) associated with memory cells within the memory devices **130**.
(17) The memory system controller **115** may include hardware such as one or more integrated circuits or discrete components, a buffer memory, or a combination thereof. The hardware may include circuitry with dedicated (e.g., hard-coded) logic to perform the operations ascribed herein to the memory system controller **115**. The memory system controller **115** may be or include a microcontroller, special purpose logic circuitry (e.g., a field programmable gate array (FPGA), an application specific integrated circuit (ASIC), a digital signal processor (DSP)), or any other suitable processor or processing circuitry.
(18) The memory system controller **115** may also include a local memory **120**. In some cases, the local memory **120** may include read-only memory (ROM) or other memory that may store operating code (e.g., executable instructions) executable by the memory system controller **115** to perform functions ascribed herein to the memory system controller **115**. In some cases, the local memory **120** may additionally or alternatively include static random access memory (SRAM) or other memory that may be used by the memory system controller **115** for internal storage or calculations, for example, related to the functions ascribed herein to the memory system controller **115**. Additionally or alternatively, the local memory **120** may serve as a cache for the memory system controller **115**. For example, data may be stored in the local memory **120** if read from or written to a memory device **130**, and the data may be available within the local memory **120** for subsequent retrieval for or manipulation (e.g., updating) by the host system **105** (e.g., with reduced latency relative to a memory device **130**) in accordance with a cache policy.
(19) Although the example of the memory system **110** in FIG. **1** has been illustrated as including the memory system controller **115**, in some cases, a memory system **110** may not include a memory system controller **115**. For example, the memory system **110** may additionally or alternatively rely upon an external controller (e.g., implemented by the host system **105**) or one or more local controllers **135**, which may be internal to memory devices **130**, respectively, to perform the functions ascribed herein to the memory system controller **115**. In general, one or more functions ascribed herein to the memory system controller **115** may in some cases instead be performed by the host system **105**, a local controller **135**, or any combination thereof. In some cases, a memory device **130** that is managed at least in part by a memory system controller **115** may be referred to as a managed memory device. An example of a managed memory device is a managed NAND (MNAND) device.
(20) A memory device **130** may include one or more arrays of non-volatile memory cells. For example, a memory device **130** may include NAND (e.g., NAND flash) memory, ROM, phase change memory (PCM), self-selecting memory, other chalcogenide-based memories, ferroelectric random access memory (FeRAM), magneto RAM (MRAM), NOR (e.g., NOR flash) memory, Spin Transfer Torque (STT)-MRAM, conductive bridging random access memory (CBRAM), resistive random access memory (RRAM), oxide based RRAM (OxRAM), electrically erasable programmable ROM (EEPROM), or any combination thereof. Additionally or alternatively, a memory device **130** may include one or more arrays of volatile memory cells. For example, a memory device **130** may include random access memory (RAM) memory cells, such as dynamic

RAM (DRAM) memory cells and synchronous DRAM (SDRAM) memory cells.
(21) In some examples, a memory device **130** may include (e.g., on a same die or within a same package) a local controller **135**, which may execute operations on one or more memory cells of the respective memory device **130**. A local controller **135** may operate in conjunction with a memory system controller **115** or may perform one or more functions ascribed herein to the memory system controller **115**. For example, as illustrated in FIG. **1**, a memory device **130**-*a* may include a local controller **135**-*a* and a memory device **130**-*b* may include a local controller **135**-*b*.
(22) In some cases, a memory device **130** may be or include a NAND device (e.g., NAND flash device). A memory device **130** may be or include a die **160** (e.g., a memory die). For example, in some cases, a memory device **130** may be a package that includes one or more dies **160**. A die **160** may, in some examples, be a piece of electronics-grade semiconductor cut from a wafer (e.g., a silicon die cut from a silicon wafer). Each die **160** may include one or more planes **165**, and each plane **165** may include a respective set of blocks **170**, where each block **170** may include a respective set of pages **175**, and each page **175** may include a set of memory cells.
(23) In some cases, planes **165** may refer to groups of blocks **170**, and in some cases, concurrent operations may take place within different planes **165**. For example, concurrent operations may be performed on memory cells within different blocks **170** so long as the different blocks **170** are in different planes **165**. In some cases, performing concurrent operations in different planes **165** may be subject to one or more restrictions, such as identical operations being performed on memory cells within different pages **175** that have the same page address within their respective planes **165** (e.g., related to command decoding, page address decoding circuitry, or other circuitry being shared across planes **165**).
(24) In some cases, a block **170** may include memory cells organized into rows (pages **175**) and columns (e.g., strings, not shown). For example, memory cells in a same page **175** may share (e.g., be coupled with) a common word line, and memory cells in a same string may share (e.g., be coupled with) a common digit line, which may alternatively be referred to as a bit line.
(25) For some NAND architectures, memory cells may be read and programmed (e.g., written) at a first level of granularity (e.g., at the page level of granularity) but may be erased at a second level of granularity (e.g., at the block level of granularity). That is, a page **175** may be the smallest unit of memory (e.g., set of memory cells) that may be independently programmed or read (e.g., programed or read concurrently as part of a single program or read operation), and a block **170** may be the smallest unit of memory (e.g., set of memory cells) that may be independently erased (e.g., erased concurrently as part of a single erase operation). Further, in some cases, NAND memory cells may be erased before they can be re-written with new data. Thus, for example, a used page **175** may in some cases not be updated until the entire block **170** that includes the page **175** has been erased.
(26) In some cases, to update some data within a block **170** while retaining other data within the block **170**, the memory device **130** may copy the data to be retained to a new block **170** and write the updated data to one or more remaining pages of the new block **170**. The memory device **130** (e.g., the local controller **135**) or the memory system controller **115** may mark or otherwise designate the data that remains in the old block **170** as invalid or obsolete and may update a logical-to-physical (L2P) mapping table to associate the logical address (e.g., logical block address) for the data with the new, valid block **170** rather than the old, invalid block **170**. In some cases, such copying and remapping may be performed instead of erasing and rewriting the entire old block **170** due to latency or wearout considerations, for example. In some cases, one or more copies of an L2P mapping table may be stored within the memory cells of the memory device **130** (e.g., within one or more blocks **170** or planes **165**) for use (e.g., reference and updating) by the local controller **135** or memory system controller **115**.
(27) In some cases, a memory system controller **115** or a local controller **135** may perform operations (e.g., as part of one or more media management algorithms) for a memory device **130**,

such as wear leveling, background refresh, garbage collection, scrub, block scans, health monitoring, or others, or any combination thereof. For example, within a memory device **130**, a block **170** may have some pages **175** containing valid data and some pages **175** containing invalid data. To avoid waiting for all of the pages **175** in the block **170** to have invalid data in order to erase and reuse the block **170**, an algorithm referred to as "garbage collection" may be invoked to allow the block **170** to be erased and released as a free block for subsequent write operations. Garbage collection may refer to a set of media management operations that include, for example, selecting a block **170** that contains valid and invalid data, selecting pages **175** in the block that contain valid data, copying the valid data from the selected pages **175** to new locations (e.g., free pages **175** in another block **170**), marking the data in the previously selected pages **175** as invalid, and erasing the selected block **170**. As a result, the quantity of blocks **170** that have been erased may be increased such that more blocks **170** are available to store subsequent data (e.g., data subsequently received from the host system **105**).

(28) In some cases (e.g., to improve reliability, to improve data retention), a memory system controller **115** or a local controller **135** may perform one or more refresh operations, which may include erasing and re-programming memory cells of a die **160** to re-establish a physical state (e.g., stored charge, polarization state, resistance state) or logical state stored in the memory cells. In some examples, refresh operations may be performed at a block level (e.g., a physical block level), where such refresh operations may include erasing and re-programming one or more blocks **170**. Refresh operations may be controlled or managed by a host system **105** (e.g., a host system controller **106**), a memory system **110** (e.g., a memory system controller **115**, a local controller **135**), or various combinations thereof, which may include refresh configuration, refresh initiation, refresh interruption, and refresh progress management. In various examples of host-initiated refresh operations, a memory system **110** may be configured to perform refresh operations requested by a host system **105** regardless of whether the memory system **110** has identified that a block **170** meets a criteria for refresh operations, or may be configured to perform refresh operations requested by a host system **105** on blocks **170** that the memory system **110** has identified as meeting a criteria for refresh operations. Additionally or alternatively, a memory system **110** may initiate refresh operations that may be transparent to the host system **105**.

(29) The system **100** may include any quantity of non-transitory computer readable media that support topology-based retirement in a memory system. For example, the host system **105**, the memory system controller **115**, or a memory device **130** may include or otherwise may access one or more non-transitory computer readable media storing instructions (e.g., firmware) for performing the functions ascribed herein to a host system **105**, a memory system controller **115**, or a memory device **130** (e.g., a local controller **135**). For example, such instructions, if executed by a host system **105** (e.g., by a host system controller **106**), by a memory system controller **115**, or by a memory device **130** (e.g., by a local controller **135**), may cause the host system **105**, the memory system controller **115**, or the memory device **130** to perform one or more associated functions as described herein.

(30) In some examples, structures of a memory device **130** (e.g., of a die **160**) may wear or degrade over time, which may lead to operational failures or otherwise unreliable operation of the memory device **130**. Some applications may not be configured to tolerate a loss or reduction of storage capacity in a memory device **130**, so a memory system **110** or a memory device **130** may be configured to detect indications of unreliable or failed access operations, and retire portions of a memory array in response to such indications. For example, a memory device **130** may be over-provisioned with spare blocks **170** (e.g., a pool of replacement blocks **170**) that provide storage beyond a stated or rated capacity of the memory device **130**, but that may not be used until they are made active or valid. As valid blocks **170** (e.g., originally-provisioned blocks **170**) wear or degrade, they may be retired by taking them out of service and replacing them with spare blocks **170** (e.g., removing a retired block **170** from a physical address space available for access operations, adding

a spare block **170** to a physical address space available for access operations).

(31) In some examples, a block retirement may include indicating that a retired block **170** is not available for access (e.g., not available for writing information, marking physical addresses associated with the retired block **170** as invalid) and indicating that a spare block **170** is available for access. In some examples, a block retirement may include removing an indication that a retired block **170** is available for access, and adding an indication that a spare block **170** is available for access. In some cases, a retirement may include or be accompanied by a transfer of information from a retired block **170** into another block **170** (e.g., an already-valid or already-available block **170**, a newly-valid or newly-available block **170**, such as a spare block **170** associated with an over-provisioning), which may include a remapping of logical addresses to new physical addresses of the other block **170**. In some examples, a retirement may include a partial or restricted access of a block **170**, such as configuring a retired block **170** for operation in a read-only mode (e.g., to support aspects of information retrieval from a retired block **170**).

(32) According to some techniques, retirement of a block **170** may be initiated in response to an error detection when accessing memory cells of the block **170**. For example, a block **170** may be retired in response to an erase failure of the block **170** (e.g., the block **170** returning an erase status fail, such as the block **170** failing an erase operation) or a program failure of the block **170** (e.g., one or more pages **175** of the block **170** returning a program status fail, such as pages **175** failing a write operation). Additionally or alternatively, a block **170** may be retired in response to a read scan failure, such as an uncorrectable error (e.g., an uncorrectable error correction code (UECC) error, an uncorrectable read error) of the block **170** (e.g., during a garbage collection operation). However, some techniques for detecting unreliable or failed access operations (e.g., failed read operations) may be overly conservative, such as being overly sensitive to transient error conditions, which may lead to excessive retirement, accelerated depletion of spare blocks **170**, or reduced life cycle of a memory device **130**, among other drawbacks or combinations thereof.

(33) In accordance with examples as disclosed herein, a memory system **110** or a memory device **130** may be configured to perform topology-based evaluations for retirement of portions of a memory array. For example, a memory system **110** (e.g., a memory system controller **115**) or a memory device **130** (e.g., a local controller **135**) may be configured to evaluate error conditions relative to a physical or electrical organization of a die **160**, or a memory array thereof, which may support inferring a presence or absence of defects (e.g., physical defects, material defects, electrical defects) in one or more structures of a memory device **130**. For example, based on various evaluations of detected errors, a memory system **110** or a memory device **130** may be able to detect or infer a presence of a short-circuit defect, an open circuit defect, a dielectric breakdown or other leakage defect, or other defects of a memory array that may be related to wear or degradation over time and, in response, may retire a portion of a memory array based on such a detection or inference. In some examples, such techniques may support distinguishing between intrinsic and extrinsic errors or failures, which may inhibit, lessen, or otherwise mitigate retirement techniques that may be overly sensitive to transient phenomena such as elevated temperatures, voltage irregularities, radiation, or other transients. Compared to other techniques of array retirement that do not consider a physical or electrical organization of a die **160**, or techniques that do not otherwise distinguish between intrinsic and extrinsic failures, implementing one or more aspects of topology-based retirement in accordance with examples as disclosed herein may support a memory device **130** being configured with a larger capacity, a smaller degree of over-provisioning, or a longer life cycle, among other beneficial configurations or combinations thereof.

(34) FIG. **2** shows a block diagram of a circuit **200** that supports topology-based retirement in a memory system in accordance with examples as disclosed herein. The circuit **200** may be included in a memory system **110**, and may include one or more components of a memory device **130**. For example, the circuit **200** illustrates an example of a block **170**-*a* including an array of memory cells **205**. Each of the memory cells **205** may be located at or otherwise accessible according to an

intersection of a word line **225** (e.g., a WL) and a bit line **235** (e.g., a BL), which may each be referred to as access lines of the block **170**-*a*. Memory cells **205** along a word line **225** may be an example of a page **175**. The word lines **225** and bit lines **235** may be coupled with a row decoder **220** and a column decoder **230**, respectively, for controlling various biasing or activation of the respective access lines. In some examples, the row decoder **220** and the column decoder **230** may be components of a local controller **135**, which may support access operations such as writing logic states to memory cells **205** or sensing logic states stored in memory cells **205**, among other operations and signaling thereof. The row decoder **220** and the column decoder **230** may be coupled with a controller **240** that is configured to perform various techniques for topology-based retirement as disclosed herein. In various examples, the controller **240** may be included in a memory system controller **115**, included in a local controller **135**, or distributed between a memory system controller **115** and a local controller **135**, among other configurations.

(35) The memory cells **205** may be physically or electrically arranged according to subblocks **210** (e.g., a first subblock **210**-*a* and a second subblock **210**-*b*). In some examples, each of the subblocks **210** may include or refer to a subset of the word lines **225** of a block **170** (e.g., subblock **210**-*a* including or associated with word lines WLa**1** through WLam, subblock **210**-*b* including or associated with word lines WLb**1** through WLbm). Although the block **170**-*a* is illustrated as including two subblocks **210**, a block **170** in accordance with the described techniques may include any quantity of subblocks **210** (e.g., two, three, four, five, six, seven, eight, etc.). Additionally or alternatively, although the subblocks **210** are illustrated as each including a respective subset of the word lines **225** of the block **170**-*a* and all of the bit lines **235** of the block **170**-*a*, in some examples, subblocks **210** may each include a respective subset of the bit lines **235** of a block **170** (e.g., in combination with including some or all of the word lines **225** of the block **170**).

(36) In some examples, one or more structures of the circuit **200** (e.g., structures of a die **160**) may wear or degrade over time, which may lead to one or more physical defects. For example, circuit **200** illustrates an example of a defect **250**-*a*, which may be associated with at least a word line WLai, and a defect **250**-*b*, which may be associated with at least a bit line BLj. The defects **250** may refer to various degradation or failure of one or more physical elements of the circuit **200**. For example, a defect **250** may refer to a short circuit defect or other dielectric breakdown (e.g., a leakage path), such as a short circuit between an access line and a chassis or ground structure of a die **160**, a short circuit between an access line and a voltage source (e.g., a positive voltage source, a negative voltage source), a short circuit between a first access line and a second access line (e.g., between a first word line **225** and a second word line **225**, between a first bit line **235** and a second bit line **235**, between a word line **225** and a bit line **235**), or between other structures of the circuit **200**. In some examples, a defect **250** may refer to an open circuit defect or other reduction or suppression of conductivity, such as a break in conductivity or a break or other cross-sectional reduction in a conductive path of an access line. The examples of defects **250**-*a* and **250**-*b* are for illustrative purposes, and a circuit may develop any quantity of one or more defects **250** in various locations of a circuit of a memory device **130**, and a defect **150** may affect any quantity of one or more access lines.

(37) In some examples, a defect **250** may cause errors when accessing memory cells **205** of the block **170**-*a*, which may include uncorrectable errors (e.g., when a quantity of errors exceeds an error correction capability of a memory system **110** or a memory device **130** that includes the circuit **200**, when an error is not due to a temporary condition). For example, if the defect **250**-*a* is an open circuit defect, the defect **250**-*a* may cause errors when accessing memory cells **205** along the word line WLai that are downstream of the defect **250**-*a* (e.g., downstream relative to the row decoder, farther from the row decoder **220** than the defect **250**-*a*) due to signals not being conveyed through the defect **250**-*a*). Likewise, if the defect **250**-*b* is an open circuit defect, the defect **250**-*b* may cause errors when accessing memory cells **205** along the bit line BLj that are downstream of the defect **250**-*b* (e.g., downstream relative to the column decoder **230**, farther from the column

decoder **230** than the defect **250-***b*) due to signals not being conveyed through the defect **250-***b*). In examples where a defect **250** is a short circuit, dielectric breakdown, or leakage defect associated with an access line, such a defect **250** may cause errors when accessing both memory cells **205** that are downstream of the defect **250** and memory cells **205** that are upstream of the defect **250** (e.g., to voltage instability or charge leakage that generally affects signaling of the access line).
(38) In some examples, a topology of the block **170-***a* may be leveraged for evaluating error conditions of the block **170-***a*. For example, the subblock **210-***a* exhibiting a higher occurrence of errors (e.g., read errors, uncorrectable errors) than the subblock **210-***b* may be indicative of a defect **250-***b* (e.g., an open circuit defect), because such a defect **250-***b* may inhibit or otherwise disrupt signaling between the column decoder **230** (e.g., and the controller **240**, and sensing circuitry coupled with the column decoder and configured to detect logic states stored by memory cells **205**) and memory cells **205** that are downstream of the defect **250-***b*, and may not affect accessing the memory cells **205** of the subblock **210-***b*. In some examples, the subblock **210-***a* exhibiting a higher occurrence of errors than the subblock **210-***b*, or the word line WLai exhibiting a higher occurrence of errors than other (e.g., neighboring, adjacent) word lines **225**, may be indicative of a defect **250-***a* (e.g., an open circuit defect or a closed circuit defect), because such a defect **250-***a* may induce failures or errors along the word line WLai (e.g., downstream of the defect **250-***a*, upstream of the defect **250-***a*), which may not affect accessing the memory cells **205** of the subblock **210-***b* or other word lines **225**. In some examples, topological comparisons of errors such as these may be additionally or alternatively performed between or among bit lines **235**, or groups of bit lines **235** (e.g., subblocks **210** each including respective sets of bit lines **235**) for various evaluations to detect or infer a presence of one or more defects **250**.
(39) Thus, in accordance with these and other examples, a comparison of quantities of errors between or among subblocks **210**, between or among word lines **225**, or between or among bit lines **235**, or other topological comparisons or combinations thereof, may be leveraged to evaluate or infer the presence of defects **250**. For example, a signature of a defect **250** may be identified or inferred based on which subblocks **210**, which word lines **225**, or which bit lines **235** demonstrate access errors (e.g., subblocks **210** or word lines **225** that are physically or electrically located relatively farther from a column decoder **230**, bit lines **235** that are located relatively farther from a row decoder **220**, or subblocks **210**, word lines **225**, or bit lines **235** having significantly higher occurrence of errors than their neighbors), or identified or inferred based on how many errors occur (e.g., how many bits are flipped, how many codewords within a word line **225** have errors), or identified or inferred based on a combination of these and other characteristics. Such topological comparisons (e.g., comparisons that consider relative physical or electrical groupings or positioning of detected errors) may improve retirement evaluations by improving insight into the cause of access errors or failures. For example, according to these and other techniques in accordance with examples as disclosed herein, such comparisons may support distinguishing between extrinsic failures (e.g., local physical issues, such as defects **250**, that may be persistent) and intrinsic failures (e.g., transient effects that may not be persistent, stochastic errors that may not have an association with a particular physical or electrical location in an array), which may avoid unnecessarily retiring blocks **170** that are affected by transient issues but may be otherwise normally operable (e.g., upon the passing or clearing of transient conditions).
(40) In some examples, retirement evaluations may be conditional or otherwise modified based on an evaluation or detection of operating conditions, such as observations of temperature, voltage, or other conditions that may affect operation of a memory system **110** or memory device **130**. For example, elevated operating temperatures may be associated with increased occurrence of access errors (e.g., appearing as a relatively substantial, localized failure), but occurrence of such access errors may decrease as operating temperature falls. Thus, retirement of a block **170** based on error detections while at an elevated operating temperature may be overly conservative, because the block **170** may perform normally or otherwise acceptably at lower temperatures. Accordingly, some

examples of topology-based retirement evaluations may include evaluations of conditions that may be transient, such as retiring a block **170** based on error evaluations if a temperature (e.g., as measured or detected by a temperature sensor **245**) is found to be within a nominal or otherwise configured range, or if a voltage (e.g., as measured or detected by a voltage sensor **246**) is found to be within a nominal or otherwise configured range, among other evaluations that consider whether operating conditions of a memory system **110** or a memory device **130** are in a normal range or are subject to transient effects that may have induced observed errors.

(41) In an example of a topology-based evaluation for retirement of a block **170**, if an error (e.g., an uncorrectable error, a UECC error) occurs during a read operation of the block **170**, a controller may initiate a retirement evaluation that includes a refresh of the block **170** or other read scan of the block **170**. During the refresh or read scan, a controller (e.g., firmware) may keep track of error counts (e.g., UECC count) per word line **225** and per subblock **210**, which may include error counts from some or all upper pages of the block **170**. After completion of at least part of the refresh or other read scan, a determination to retire the block **170** may be based on various error criteria and operating condition criteria. For example, the block **170** may be retired if a quantity of errors for some quantity of word lines **225** (e.g., one word line, two word lines) is greater than or equal to four errors, or some other threshold (e.g., a variable that may be configured during a device qualification), while other word lines **225** have zero errors. Additionally or alternatively, the block **170** may be retired if a quantity of errors for a subblock **210** is greater than a threshold (e.g., a configured parameter, or a variable that is configured during a device qualification, or a combination thereof), while other subblocks **210** have zero errors. In some examples, either or both of such retirement determinations may be conditional based on operating temperature, such as retirement being conditional on a difference between write temperature and read temperature being less than a temperature threshold. In some examples, if a read temperature is not outside a nominal or configured range (e.g., within a range defined by a configured lower temperature and a configured upper temperature), the block **170** may be retired, and if a read temperature is outside the nominal or configured range, the block **170** may not be retired (e.g., due to an elevated probability that observed errors are related to a transient effect). In some examples, such a retirement determination may also be conditional on the availability of spare blocks **170** (e.g., of an over-provisioning, of a pool of spare blocks **170**).

(42) FIG. **3** shows a process flow illustrating a method **300** that supports topology-based retirement in a memory system in accordance with examples as disclosed herein. Aspects of the method **300** may be implemented by a controller (e.g., a memory system controller **115**, a local controller **135**, a controller **240**), among other components. Additionally or alternatively, aspects of the method **300** may be implemented as instructions stored in memory (e.g., firmware stored in a memory coupled with a memory system **110** or a memory device **130**). For example, the instructions, if executed by a controller, may cause the controller to perform the operations of the method **300**.

(43) In some examples, the method **300** may begin (e.g., be initiated) according to a condition or criteria being satisfied. For example, the method **300** may begin upon a determination of an error condition, such as an uncorrectable read error (e.g., of a block **170**), which may be determined during the course of normal operations (e.g., during a read operation performed in response to a read command from a host system **105**), media management operations (e.g., a garbage collection operation, a wear leveling operation), or other operations. In some examples, the method **300** may begin according to an initiation interval, such as satisfying a threshold duration (e.g., a data retention duration) or a threshold quantity of access operations (e.g., read operations, since a power cycle, since an installation in a user product, since a prior retirement evaluation such as method **300**), among other intervals. In some examples, the method **300** may begin if an operating temperature satisfies a threshold, which may or may not be combined with other initiation criteria (e.g., a determination of an error condition while operating in a configured temperature range, upon satisfying an initiation interval while operating in a configured temperature range).

(44) At **310**, the method **300** may include performing a read scan of the memory array, such as a refresh operation, a media scan, a media management operation, or some other scanning operation of a block **170** that was identified as having an uncorrectable error (e.g., a UECC error) or a block that was otherwise identified for a retirement evaluation. In some examples, a refresh of the block **170** at **310** may include moving the information stored in the block **170** to another block **170**, which accordingly may include reading the whole block **170** (e.g., pages **175** of the block). Other examples of a read scan at **310** may or may not include a refresh operation. For example, the method **300** may additionally or alternatively include other types of read scans that include reading a block **170** (e.g., each page **175** of a block **170**) for a topology-based retirement evaluation, which may or may not include moving information or otherwise performing write operations associated with data stored in the block **170**.

(45) As part of or in parallel with the read scan of the block **170** (e.g., while reading the block **170** associated with the uncorrectable error), the method **300** may include tracking a quantity of errors (e.g., errors identified while performing the read scan of **310**, read errors, uncorrectable errors, UECC errors) in accordance with a topology of the block **170**. For example, the method **300** may include tracking a quantity of errors per word line **225** and a quantity of errors per subblock **210**. Although the method **300** is described in the context of errors per subblock **210**, in some examples, the method **300** may be modified by additionally or alternatively tracking a quantity of errors per bit line **235** or groups of bit lines **235**, which may support finer granularity for retirement evaluations. In various examples, the method **300** may proceed to **320** after the read scan operations of **310** are partially or fully complete.

(46) At **320**, the method **300** may include determining whether a quantity of errors on one or more WLs satisfies (e.g., is greater than, is greater than or equal to) a WL error threshold. In some examples, an evaluation of **320** may include a stepwise or otherwise individual comparison of respective errors for one or more of the WLs of the block **170** (e.g., whether a quantity or errors on a first WL **225** is greater than a threshold, whether a quantity of errors on a second WL **225** is greater than a threshold, and so on). In some examples, an evaluation of **320** may include a determination of whether some WLs **225** (e.g., WLs that are relatively closer to a column decoder **230**) are associated with a quantity of errors that is less than or otherwise satisfies a first threshold and other WLs **225** (e.g., WLs that are relatively farther from a column decoder **230**) are associated with a quantity of errors that is greater than or otherwise satisfies a second threshold. WL error thresholds may be a fixed or otherwise configured value (e.g., one error, four errors, or any other quantity of errors), which may be programmed as part of a manufacturing operation. In some examples, the WL error threshold may be a relative value, such as an average quantity of errors per WL **225** (e.g., for WLs other than a WL being evaluated, for all WLs of the block), or a quantity of errors for one or more neighboring WLs **225** (e.g., one or more WLs directly adjacent to a WL being evaluated, one or more sets of WLs directly adjacent to a WL being evaluated), among other examples of comparisons between quantities of errors associated with different WLs **225**. In some examples, an evaluation at **320** may include a determination of whether a quantity of UECC errors (e.g., on a defined page type) on some quantity of one or more WLs **225** is greater than a WL UECC threshold. If, at **320**, a quantity of errors on one or more WLs **225** satisfies the WL error threshold (e.g., indicating error conditions that may be indicative of a defect **250**), the method **300** may proceed to **325**. If, at **320**, a quantity of errors on one or more WLs **225** does not satisfy the WL error threshold (e.g., indicating error conditions that may not be indicative of a defect **250**), the method **300** may proceed to **330**.

(47) At **325**, the method **300** may include determining whether a temperature of the memory system **110** (e.g., of the memory device **130**, of the die **160**) satisfies (e.g., is less than, is less than or equal to) a threshold. In some examples, an evaluation at **325** may include a determination of whether a current temperature satisfies a threshold, or whether a temperature during the read scan operation of **310** satisfies a threshold. In some examples, an evaluation at **325** may include a

determination of whether an absolute difference between a read temperature (e.g., a temperature associated with reading the memory block **170**, a temperature associated with the read scan operations of **310**) and a write temperature (e.g., a temperature associated with the writing of data to the memory block **170**) is less than a threshold temperature difference.
(48) If, at **325**, the temperature of the memory system **110** satisfies the threshold (e.g., indicating a relatively higher likelihood of tracked WL errors being associated with a defect **250**, indicating a relatively lower likelihood of tracked WL errors being associated with transients), the method **300** may proceed to **340**. If, at **325**, the temperature of the memory system **110** does not satisfy the threshold (e.g., indicating a relatively lower likelihood of tracked WL errors being associated with a defect, indicating a relatively higher likelihood of tracked WL errors being associated with transients), the method **300** may proceed to **330**.
(49) At **330**, the method **300** may include determining whether a quantity of errors on one or more subblocks **210** satisfies (e.g., is greater than, is greater than or equal to) a subblock error threshold. In some examples, an evaluation of **330** may include a stepwise or otherwise individual comparison of respective errors for one or more of the subblocks **210** of the block **170** (e.g., whether a quantity or errors on a first subblock is greater than a threshold, whether a quantity of errors on a second subblock is greater than a threshold, and so on). In some examples, an evaluation of **330** may include a determination of whether some subblocks **210** (e.g., subblocks that are relatively closer to a row decoder **220**) are associated with a quantity of errors that is less than or otherwise satisfies a first threshold and other subblocks **210** (e.g., subblocks that are relatively farther from a row decoder **220**) are associated with a quantity of errors that is greater than or otherwise satisfies a second threshold. Subblock error thresholds may be a fixed or otherwise configured value, which may be programmed as part of a manufacturing operation. In some examples, the subblock error threshold may be a relative value, such as an average quantity of errors per subblock **210** (e.g., for subblocks other than a subblock being evaluated, for some or all subblocks of the block), or a quantity of errors for one or more neighboring subblocks **210** (e.g., one or more subblocks directly adjacent to a subblock being evaluated, one or more sets of subblocks directly adjacent to a subblock being evaluated), among other examples of comparisons between quantities of errors associated with different subblocks **210**. In some examples, an evaluation at **330** may include a determination of whether a quantity of UECC errors (e.g., on a defined page type) on any one or more subblocks **210** is greater than a subblock threshold. If, at **330**, a quantity of errors on one or more subblocks **210** satisfies the subblock error threshold (e.g., indicating error conditions that may be indicative of a defect **250**), the method **300** may proceed to **335**. If, at **330**, a quantity of errors on one or more subblocks **210** (e.g., some or all subblocks) does not satisfy the subblock error threshold (e.g., indicating error conditions that may not be indicative of a defect **250**), the method **300** may proceed to **345**. Although the operations of **330** are shown as being after the operations of **320**, in some examples, the relative position may be reversed.
(50) At **335**, the method **300** may include determining whether a temperature of the memory system **110** (e.g., of the memory device **130**, of the die **160**) satisfies (e.g., is less than, is less than or equal to) a threshold, which may or may not be a same evaluation or a similar evaluation as **325** (e.g., using a same or different temperature threshold, using a same or different reference temperature). If, at **335**, the temperature of the memory system **110** satisfies the threshold (e.g., indicating a relatively higher likelihood of tracked subblock errors being associated with a defect **250**, indicating a relatively lower likelihood of tracked subblock errors being associated with transients), the method **300** may proceed to **340**. If, at **335**, the temperature of the memory system **110** does not satisfy the threshold (e.g., indicating a relatively lower likelihood of tracked subblock errors being associated with a defect, indicating a relatively higher likelihood of tracked subblock errors being associated with transients), the method **300** may proceed to **345**.
(51) At **340**, the method **300** may include retiring the block **170**. In various examples, retiring the block **170** may include setting an indication that the block **170** is generally unavailable for access

operations or unavailable for write operations. In some examples, the block **170** may still be operable in a read-only configuration. The physical addresses of the block **170** may be replaced by physical addresses of a new block **170** (e.g., from a replacement pool), which may be remapped to logical addresses in an L2P table. At **345**, the method **300** may include not retiring the subblock, and the method may proceed to **350**.
(52) At **350**, the method **300** may end. In various examples, the ending of the method **300** may or may not include an indication that the method **300** has ended. In some examples, the ending of the method **300** may include an indication that the read scan of **310** has ended. In some examples, the ending of the method **300** may include a reallocation or idling of resources (e.g., processing resources, cache resources) that were used during the method **300**.
(53) By applying these and other techniques for topology-based retirement in accordance with examples as disclosed herein, a memory system **110** may be provided with improved performance compared to other techniques that do not include topology-based evaluations for retirement. For example, the described techniques may reduce a probability of retiring blocks **170** with errors that were related to transients, but are otherwise still operable (e.g., allowing transient failures and intrinsic issues to be ignored). In some examples, implementing such techniques may reduce a quantity of spare blocks **170** (e.g., of a replacement pool) or otherwise reduce a degree of overprovisioning in a memory system **110** for a given life cycle. Moreover, in some examples, a transition to a read-only mode may be delayed, or a life cycle of a memory system **110** may otherwise be lengthened due to more efficient block usage (e.g., more efficient use of over-provisioning) for a given quantity of spare blocks **170**.
(54) FIG. **4** shows a block diagram **400** of a memory system **420** that supports topology-based retirement in a memory system in accordance with examples as disclosed herein. The memory system **420** may be an example of aspects of a memory system as described with reference to FIGS. **1** through **3** (e.g., a memory system **110**). The memory system **420**, or various components thereof (e.g., a memory system controller **115**, a local controller **135**), may be an example of means for performing various aspects of topology-based retirement in a memory system as described herein. For example, the memory system **420** may include an error evaluation component **425**, a block retirement component **430**, an operating mode management component **435**, an information transfer component **440**, a temperature evaluation component **445**, a read scan component **450**, or any combination thereof. Each of these components may communicate, directly or indirectly, with one another (e.g., via one or more buses).
(55) The error evaluation component **425** may be configured as or otherwise support a means for determining a first quantity of access errors for a first subblock of a block of memory cells of the memory system **420**. In some examples, the error evaluation component **425** may be configured as or otherwise support a means for determining a second quantity of access errors for a second subblock of the block of memory cells. In some examples, the block retirement component **430** may be configured as or otherwise support a means for retiring the block of memory cells based at least in part on (e.g., in response to, as initiated by, after meeting a condition of) a comparison between the first quantity of access errors and the second quantity of access errors.
(56) In some examples, to support retiring the block of memory cells, the block retirement component **430** may be configured as or otherwise support a means for retiring the block of memory cells based at least in part on (e.g., in response to, as initiated by, after meeting a condition of) a difference between the second quantity of access errors and the first quantity of access errors satisfying a threshold (e.g., meeting or exceeding the threshold).
(57) In some examples, to support retiring the block of memory cells, the block retirement component **430** may be configured as or otherwise support a means for retiring the block of memory cells based at least in part on (e.g., in response to, as initiated by, after meeting a condition of) the first quantity of access errors satisfying a first threshold (e.g., meeting or being below the first threshold) and the second quantity of access errors satisfying a second threshold (e.g., meeting

or exceeding the second threshold).
(58) In some examples, the operating mode management component **435** may be configured as or otherwise support a means for operating the block of memory cells in a read-only mode based at least in part on (e.g., in response to, as initiated by, after meeting a condition of) the retiring.
(59) In some examples, to support retiring the block of memory cells, the block retirement component **430** may be configured as or otherwise support a means for indicating that the block of memory cells is unavailable for writing information, or a means for removing an indication that the block of memory cells is available for writing, or a combination thereof.
(60) In some examples, to support retiring the block of memory cells, the information transfer component **440** may be configured as or otherwise support a means for transferring information stored in the block of memory cells to a second block of memory cells of the memory system **420**.
(61) In some examples, the temperature evaluation component **445** may be configured as or otherwise support a means for determining an operating temperature of the memory system **420** associated with the first quantity of access errors and the second quantity of access errors. In some examples, to support retiring the block of memory cells, the block retirement component **430** may be configured as or otherwise support a means for retiring the block of memory cells based at least in part on (e.g., in response to, as initiated by, after meeting a condition of) the comparison between the first quantity of access errors and the second quantity of access errors and the operating temperature satisfying a temperature threshold (e.g., meeting or being below a temperature threshold).
(62) In some examples, the error evaluation component **425** may be configured as or otherwise support a means for identifying an uncorrectable error based at least in part on (e.g., during, as part of, in response to) accessing the block of memory cells. In some examples, the read scan component **450** may be configured as or otherwise support a means for initiating a read scan operation (e.g., a refresh operation or portion thereof, a media management operation that includes a read scan) on the block of memory cells based at least in part on (e.g., in response to, as initiated by, after meeting a condition of) the error evaluation component **425** identifying the uncorrectable error. In some examples, the error evaluation component **425** may be configured as or otherwise support a means for determining the first quantity of access errors and the second quantity of access errors based at least in part on the read scan component **450** initiating the read scan operation (e.g., during the read scan operation, as part of the read scan operation, in response to the read scan operation).
(63) In some examples, to support determining the first quantity of access errors, the error evaluation component **425** may be configured as or otherwise support a means for determining a first quantity of errors associated with a word line of the first subblock. In some examples, to support determining the second quantity of access errors, the error evaluation component **425** may be configured as or otherwise support a means for determining a second quantity of errors associated with a word line of the second subblock.
(64) In some examples, to support determining the first quantity of access errors, the error evaluation component **425** may be configured as or otherwise support a means for determining a first total quantity of errors associated with the first subblock. In some examples, to support determining the second quantity of access errors, the error evaluation component **425** may be configured as or otherwise support a means for determining a second total quantity of errors associated with the second subblock.
(65) In some examples, to support determining the first quantity of access errors, the error evaluation component **425** may be configured as or otherwise support a means for determining a first quantity of uncorrectable errors associated with reading memory cells of the first subblock. In some examples, to support determining the second quantity of access errors, the error evaluation component **425** may be configured as or otherwise support a means for determining a second quantity of uncorrectable errors associated with reading memory cells of the second subblock.

(66) In some examples, the first subblock and the second subblock may share a plurality of access lines (e.g., bit lines, which may be common from the first subblock to the second subblock).
(67) FIG. **5** shows a flowchart illustrating a method **500** that supports topology-based retirement in a memory system in accordance with examples as disclosed herein. The operations of method **500** may be implemented by a memory system or its components as described herein. For example, the operations of method **500** may be performed by a memory system as described with reference to FIGS. **1** through **4** (e.g., a memory system **110**, a memory system controller **115**, a local controller **135**, a memory system **420**). In some examples, a memory system, or component thereof, may execute a set of instructions to control the functional elements of the device to perform the described functions. Additionally or alternatively, the memory system may perform aspects of the described functions using special-purpose hardware.
(68) At **505**, the method may include determining a first quantity of access errors for a first subblock of a block of memory cells of the memory system. The operations of **505** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **505** may be performed by an error evaluation component **425** as described with reference to FIG. **4**.
(69) At **510**, the method may include determining a second quantity of access errors for a second subblock of the block of memory cells. The operations of **510** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **510** may be performed by an error evaluation component **425** as described with reference to FIG. **4**.
(70) At **515**, the method may include retiring the block of memory cells based at least in part on a comparison between the first quantity of access errors and the second quantity of access errors. The operations of **515** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **515** may be performed by a block retirement component **430** as described with reference to FIG. **4**.
(71) In some examples, an apparatus as described herein may perform a method or methods, such as the method **500**. The apparatus may include, features, circuitry, logic, means, or instructions (e.g., a non-transitory computer-readable medium storing instructions executable by a processor) for determining a first quantity of access errors for a first subblock of a block of memory cells of the memory system, determining a second quantity of access errors for a second subblock of the block of memory cells, and retiring the block of memory cells based at least in part on a comparison between the first quantity of access errors and the second quantity of access errors.
(72) In some examples of the method **500** and the apparatus described herein, retiring the block of memory cells may include operations, features, circuitry, logic, means, or instructions for retiring the block of memory cells based at least in part on a difference between the second quantity of access errors and the first quantity of access errors satisfying a threshold.
(73) In some examples of the method **500** and the apparatus described herein, retiring the block of memory cells may include operations, features, circuitry, logic, means, or instructions for retiring the block of memory cells based at least in part on the first quantity of access errors satisfying a first threshold and the second quantity of access errors satisfying a second threshold.
(74) Some examples of the method **500** and the apparatus described herein may further include operations, features, circuitry, logic, means, or instructions for operating the block of memory cells in a read-only mode based at least in part on the retiring.
(75) In some examples of the method **500** and the apparatus described herein, retiring the block of memory cells may include operations, features, circuitry, logic, means, or instructions for indicating that the block of memory cells is unavailable for writing information, or removing an indication that the block of memory cells is available for writing, or a combination thereof.
(76) In some examples of the method **500** and the apparatus described herein, retiring the block of memory cells may include operations, features, circuitry, logic, means, or instructions for transferring information stored in the block of memory cells to a second block of memory cells of

the memory system.
(77) Some examples of the method **500** and the apparatus described herein may further include operations, features, circuitry, logic, means, or instructions for determining an operating temperature of the memory system associated with the first quantity of access errors and the second quantity of access errors, and retiring the block of memory cells based at least in part on the comparison between the first quantity of access errors and the second quantity of access errors and the operating temperature satisfying a temperature threshold.
(78) Some examples of the method **500** and the apparatus described herein may further include operations, features, circuitry, logic, means, or instructions for identifying an uncorrectable error based at least in part on accessing the block of memory cells, initiating a read scan operation (e.g., a refresh operation or portion thereof, a media management operation that includes a read scan) on the block of memory cells based at least in part on identifying the uncorrectable error, and determining the first quantity of access errors and the second quantity of access errors based at least in part on initiating the read scan operation (e.g., as part of or in response to accessing memory cells during the read scan operation).
(79) In some examples of the method **500** and the apparatus described herein, determining the first quantity of access errors may include operations, features, circuitry, logic, means, or instructions for determining a first quantity of errors associated with a word line of the first subblock. In some examples of the method **500** and the apparatus described herein, determining the second quantity of access errors may include operations, features, circuitry, logic, means, or instructions for determining a second quantity of errors associated with a word line of the second subblock.
(80) In some examples of the method **500** and the apparatus described herein, determining the first quantity of access errors may include operations, features, circuitry, logic, means, or instructions for determining a first total quantity of errors associated with the first subblock. In some examples of the method **500** and the apparatus described herein, determining the second quantity of access errors may include operations, features, circuitry, logic, means, or instructions for determining a second total quantity of errors associated with the second subblock.
(81) In some examples of the method **500** and the apparatus described herein, determining the first quantity of access errors may include operations, features, circuitry, logic, means, or instructions for determining a first quantity of uncorrectable errors associated with reading memory cells of the first subblock. In some examples of the method **500** and the apparatus described herein, determining the second quantity of access errors may include operations, features, circuitry, logic, means, or instructions for determining a second quantity of uncorrectable errors associated with reading memory cells of the second subblock.
(82) In some examples of the method **500** and the apparatus described herein, the first subblock and the second subblock share a plurality of access lines (e.g., bit lines).
(83) It should be noted that the methods described above describe possible implementations, and that the operations and the steps may be rearranged or otherwise modified and that other implementations are possible. Further, portions from two or more of the methods may be combined.
(84) An apparatus is described. The apparatus may include a memory device including a block of memory cells and a controller coupled with the memory device (e.g., a controller in communication with the memory device, a controller of the memory device, a memory system controller **115**, a local controller **135**). The controller may be configured to cause the apparatus to determine a first quantity of access errors for a first subblock of the block of memory cells, determine a second quantity of access errors for a second subblock of the block of memory cells, and retire the block of memory cells based at least in part on a comparison between the first quantity of access errors and the second quantity of access errors.
(85) In some examples, the controller may be configured to cause the apparatus to retire the block of memory cells based at least in part on a difference between the second quantity of access errors

and the first quantity of access errors satisfying a threshold.
(86) In some examples, the controller may be configured to cause the apparatus to retire the block of memory cells based at least in part on the first quantity of access errors satisfying a first threshold and the second quantity of access errors satisfying a second threshold.
(87) In some examples, the controller may be configured to cause the apparatus to operate the block of memory cells in a read-only mode based at least in part on the retiring.
(88) In some examples, to retire the block or memory cells, the controller may be configured to cause the apparatus to indicate that the block of memory cells is unavailable for writing information, or remove an indication that the block of memory cells is available for writing, or a combination thereof.
(89) In some examples, to retire the block or memory cells, the controller may be configured to cause the apparatus to transfer information stored in the block of memory cells to a second block of memory cells of the memory device.
(90) In some examples, the controller may be configured to cause the apparatus to determine an operating temperature of the memory device associated with the first quantity of access errors and the second quantity of access errors, and retire the block of memory cells based at least in part on the comparison between the first quantity of access errors and the second quantity of access errors and the operating temperature satisfying a temperature threshold.
(91) In some examples, the controller may be configured to cause the apparatus to identify an uncorrectable error based at least in part on accessing the block of memory cells, initiate a read scan operation (e.g., a refresh operation or portion thereof, a media management operation that includes a read scan) on the block of memory cells based at least in part on identifying the uncorrectable error, and determine the first quantity of access errors and the second quantity of access errors based at least in part on initiating the read scan operation.
(92) In some examples of the apparatus, to determine the first quantity of access errors, the controller may be configured to cause the apparatus to determine a first quantity of errors associated with a word line of the first subblock and, to determine the second quantity of access errors, the controller may be configured to cause the apparatus to determine a second quantity of errors associated with a word line of the second subblock.
(93) In some examples of the apparatus, to determine the first quantity of access errors, the controller may be configured to cause the apparatus to determine a first total quantity of errors associated with the first subblock and, to determine the second quantity of access errors, the controller may be configured to cause the apparatus to determine a second total quantity of errors associated with the second subblock.
(94) In some examples of the apparatus, to determine the first quantity of access errors, the controller may be configured to cause the apparatus to determine a first quantity of uncorrectable errors associated with reading memory cells of the first subblock and, to determine the second quantity of access errors, the controller may be configured to cause the apparatus to determine a second quantity of uncorrectable errors associated with reading memory cells of the second subblock.
(95) In some examples of the apparatus, the first subblock and the second subblock share a plurality of bit lines.
(96) A non-transitory computer-readable medium is described. The non-transitory computer-readable medium may store code including instructions which, when executed by a processor of an electronic device, cause the electronic device to determine a first quantity of access errors for a first subblock of a block of memory cells of the electronic device, determine a second quantity of access errors for a second subblock of the block of memory cells, and retire the block of memory cells based at least in part on a comparison between the first quantity of access errors and the second quantity of access errors.
(97) In some examples of the non-transitory computer-readable medium, the instructions to retire

the block of memory cells, when executed by the processor of the electronic device, may cause the electronic device to retire the block of memory cells based at least in part on a difference between the second quantity of access errors and the first quantity of access errors satisfying a threshold.
(98) In some examples of the non-transitory computer-readable medium, the instructions to retire the block of memory cells, when executed by the processor of the electronic device, may cause the electronic device to retire the block of memory cells based at least in part on the first quantity of access errors satisfying a first threshold and the second quantity of access errors satisfying a second threshold.
(99) In some examples of the non-transitory computer-readable medium, the instructions, when executed by the processor of the electronic device, may cause the electronic device to operate the block of memory cells in a read-only mode based at least in part on the retiring.
(100) In some examples of the non-transitory computer-readable medium, the instructions to retire the block of memory cells, when executed by the processor of the electronic device, may cause the electronic device to indicate that the block of memory cells is unavailable for writing information, or remove an indication that the block of memory cells is available for writing, or a combination thereof.
(101) In some examples of the non-transitory computer-readable medium, the instructions to retire the block of memory cells, when executed by the processor of the electronic device, may cause the electronic device to transfer information stored in the block of memory cells to a second block of memory cells of the electronic device.
(102) In some examples of the non-transitory computer-readable medium, the instructions, when executed by the processor of the electronic device, may cause the electronic device to determine an operating temperature of the electronic device associated with the first quantity of access errors and the second quantity of access errors, and retire the block of memory cells based at least in part on the comparison between the first quantity of access errors and the second quantity of access errors and the operating temperature satisfying a temperature threshold.
(103) In some examples of the non-transitory computer-readable medium, the instructions, when executed by the processor of the electronic device, may cause the electronic device to identify an uncorrectable error based at least in part on accessing the block of memory cells, initiate a read scan operation (e.g., a refresh operation or portion thereof, a media management operation that includes a read scan) on the block of memory cells based at least in part on identifying the uncorrectable error, and determine the first quantity of access errors and the second quantity of access errors based at least in part on initiating the read scan operation.
(104) In some examples of the non-transitory computer-readable medium, the instructions to determine the first quantity of access errors, when executed by the processor of the electronic device, may cause the electronic device to determine a first quantity of errors associated with a word line of the first subblock, and the instructions to determine the second quantity of access errors, when executed by the processor of the electronic device, may cause the electronic device to determine a second quantity of errors associated with a word line of the second subblock.
(105) In some examples of the non-transitory computer-readable medium, the instructions to determine the first quantity of access errors, when executed by the processor of the electronic device, may cause the electronic device to determine a first total quantity of errors associated with the first subblock, and the instructions to determine the second quantity of access errors, when executed by the processor of the electronic device, may cause the electronic device to determine a second total quantity of errors associated with the second subblock.
(106) In some examples of the non-transitory computer-readable medium, the instructions to determine the first quantity of access errors, when executed by the processor of the electronic device, may cause the electronic device to determine a first quantity of uncorrectable errors associated with reading memory cells of the first subblock, and the instructions to determine the second quantity of access errors, when executed by the processor of the electronic device, may

cause the electronic device to determine a second quantity of uncorrectable errors associated with reading memory cells of the second subblock.

(107) In some examples of the non-transitory computer-readable medium, the first subblock and the second subblock share a plurality of access lines (e.g., bit lines).

(108) Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof. Some drawings may illustrate signals as a single signal; however, the signal may represent a bus of signals, where the bus may have a variety of bit widths.

(109) The terms “electronic communication,” “conductive contact,” “connected,” and “coupled” may refer to a relationship between components that supports the flow of signals between the components. Components are considered in electronic communication with (or in conductive contact with or connected with or coupled with) one another if there is any conductive path between the components that can, at any time, support the flow of signals between the components. At any given time, the conductive path between components that are in electronic communication with each other (or in conductive contact with or connected with or coupled with) may be an open circuit or a closed circuit based on the operation of the device that includes the connected components. The conductive path between connected components may be a direct conductive path between the components or the conductive path between connected components may be an indirect conductive path that may include intermediate components, such as switches, transistors, or other components. In some examples, the flow of signals between the connected components may be interrupted for a time, for example, using one or more intermediate components such as switches or transistors.

(110) The term “coupling” refers to a condition of moving from an open-circuit relationship between components in which signals are not presently capable of being communicated between the components over a conductive path to a closed-circuit relationship between components in which signals are capable of being communicated between components over the conductive path. If a component, such as a controller, couples other components together, the component initiates a change that allows signals to flow between the other components over a conductive path that previously did not permit signals to flow.

(111) The term “isolated” refers to a relationship between components in which signals are not presently capable of flowing between the components. Components are isolated from each other if there is an open circuit between them. For example, two components separated by a switch that is positioned between the components are isolated from each other if the switch is open. If a controller isolates two components, the controller affects a change that prevents signals from flowing between the components using a conductive path that previously permitted signals to flow.

(112) The terms “if,” “when,” “based on,” or “based at least in part on” may be used interchangeably. In some examples, if the terms “if,” “when,” “based on,” or “based at least in part on” are used to describe a conditional action, a conditional process, or connection between portions of a process, the terms may be interchangeable.

(113) The term “in response to” may refer to one condition or action occurring at least partially, if not fully, as a result of a previous condition or action. For example, a first condition or action may be performed and second condition or action may at least partially occur as a result of the previous condition or action occurring (whether directly after or after one or more other intermediate conditions or actions occurring after the first condition or action).

(114) Additionally, the terms “directly in response to” or “in direct response to” may refer to one condition or action occurring as a direct result of a previous condition or action. In some examples, a first condition or action may be performed and second condition or action may occur directly as a result of the previous condition or action occurring independent of whether other conditions or

actions occur. In some examples, a first condition or action may be performed and second condition or action may occur directly as a result of the previous condition or action occurring, such that no other intermediate conditions or actions occur between the earlier condition or action and the second condition or action or a limited quantity of one or more intermediate steps or actions occur between the earlier condition or action and the second condition or action. Any condition or action described herein as being performed “based on,” “based at least in part on,” or “in response to” some other step, action, event, or condition may additionally or alternatively (e.g., in an alternative example) be performed “in direct response to” or “directly in response to” such other condition or action unless otherwise specified.

(115) The devices discussed herein, including a memory array, may be formed on a semiconductor substrate, such as silicon, germanium, silicon-germanium alloy, gallium arsenide, gallium nitride, etc. In some examples, the substrate is a semiconductor wafer. In some other examples, the substrate may be a silicon-on-insulator (SOI) substrate, such as silicon-on-glass (SOG) or silicon-on-sapphire (SOP), or epitaxial layers of semiconductor materials on another substrate. The conductivity of the substrate, or sub-regions of the substrate, may be controlled through doping using various chemical species including, but not limited to, phosphorous, boron, or arsenic. Doping may be performed during the initial formation or growth of the substrate, by ion-implantation, or by any other doping means.

(116) A switching component or a transistor discussed herein may represent a field-effect transistor (FET) and comprise a three terminal device including a source, drain, and gate. The terminals may be connected to other electronic elements through conductive materials, e.g., metals. The source and drain may be conductive and may comprise a heavily-doped, e.g., degenerate, semiconductor region. The source and drain may be separated by a lightly-doped semiconductor region or channel. If the channel is n-type (i.e., majority carriers are electrons), then the FET may be referred to as an n-type FET. If the channel is p-type (i.e., majority carriers are holes), then the FET may be referred to as a p-type FET. The channel may be capped by an insulating gate oxide. The channel conductivity may be controlled by applying a voltage to the gate. For example, applying a positive voltage or negative voltage to an n-type FET or a p-type FET, respectively, may result in the channel becoming conductive. A transistor may be “on” or “activated” if a voltage greater than or equal to the transistor's threshold voltage is applied to the transistor gate. The transistor may be “off” or “deactivated” if a voltage less than the transistor's threshold voltage is applied to the transistor gate.

(117) The description set forth herein, in connection with the appended drawings, describes example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term “exemplary” used herein means “serving as an example, instance, or illustration” and not “preferred” or “advantageous over other examples.” The detailed description includes specific details to providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, well-known structures and devices are shown in block diagram form to avoid obscuring the concepts of the described examples.

(118) In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a hyphen and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.

(119) The functions described herein may be implemented in hardware, software executed by a processor, firmware, or any combination thereof. If implemented in software executed by a processor, the functions may be stored on or transmitted over, as one or more instructions or code, a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described

above can be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations.

(120) For example, the various illustrative blocks and components described in connection with the disclosure herein may be implemented or performed with a general-purpose processor, a DSP, an ASIC, an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any processor, controller, microcontroller, or state machine. A processor may be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

(121) As used herein, including in the claims, "or" as used in a list of items (for example, a list of items prefaced by a phrase such as "at least one of" or "one or more of") indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase "based on" shall not be construed as a reference to a closed set of conditions. For example, an exemplary step that is described as "based on condition A" may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase "based on" shall be construed in the same manner as the phrase "based at least in part on."

(122) Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A non-transitory storage medium may be any available medium that can be accessed by a general purpose or special purpose computer. By way of example, and not limitation, non-transitory computer-readable media can comprise RAM, ROM, electrically erasable programmable read-only memory (EEPROM), compact disk (CD) ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other non-transitory medium that can be used to carry or store desired program code means in the form of instructions or data structures and that can be accessed by a general-purpose or special-purpose computer, or a general-purpose or special-purpose processor. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. Disk and disc, as used herein, include CD, laser disc, optical disc, digital versatile disc (DVD), floppy disk, and Blu-ray disc, where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above are also included within the scope of computer-readable media.

(123) The description herein is provided to enable a person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

Claims

1. A memory system, comprising: a plurality of blocks of memory cells; and one or more controllers configured to cause the memory system to: determine a first quantity of errors

associated with a first subset of memory cells of a block of memory cells of the plurality of blocks; determine a second quantity of errors associated with a second subset of memory cells of the block of memory cells; and retire the block of memory cells based at least in part on a difference between the first quantity of errors and the second quantity of errors and an operating temperature of the memory system associated with the first quantity of errors and the second quantity of errors satisfying a temperature threshold.

2. The memory system of claim 1, wherein the block is associated with a plurality of word lines, and the one or more controllers are configured to cause the memory system to: determine the first quantity of errors associated with a first subset of one or more of the plurality of word lines; and determine the second quantity of errors associated with a second subset of one or more of the plurality of word lines.

3. The memory system of claim 1, wherein the block is associated with a plurality of bit lines, and the one or more controllers are configured to cause the memory system to: determine the first quantity of errors associated with a first subset of one or more of the plurality of bit lines; and determine the second quantity of errors associated with a second subset of one or more of the plurality of bit lines.

4. The memory system of claim 1, wherein the block is associated with a plurality of word lines and a plurality of bit lines, and the one or more controllers are configured to cause the memory system to: determine the first quantity of errors associated with a first subset of one or more of the plurality of word lines and a first subset of one or more of the plurality of bit lines; and determine the second quantity of errors associated with a second subset of one or more of the plurality of word lines and a second subset of one or more of the plurality of bit lines.

5. The memory system of claim 1, wherein the one or more controllers are configured to cause the memory system to: retire the block of memory cells based at least in part on the difference between the first quantity of errors and the second quantity of errors and a difference between a write temperature and a read temperature associated with the first quantity of errors and the second quantity of errors satisfying a threshold temperature difference.

6. The memory system of claim 1, wherein the one or more controllers are configured to cause the memory system to: initiate a read scan of the block of memory cells based at least in part on an identified error condition of the block of memory cells; and determine the first quantity of errors and the second quantity of errors based at least in part on the read scan of the block of memory cells.

7. The memory system of claim 1, wherein the one or more controllers are configured to cause the memory system to: retire the block of memory cells based at least in part on an availability of a second block of memory cells of the plurality of blocks; and replace one or more physical addresses of the block of memory cells with one or more physical addresses of the second block of memory cells based at least in part on the retirement.

8. A memory system, comprising: a plurality of blocks of memory cells; and one or more controllers configured to cause the memory system to: determine a first quantity of errors associated with a first subset of memory cells of a block of memory cells of the plurality of blocks; determine a second quantity of errors associated with a second subset of memory cells of the block of memory cells; and retire the block of memory cells based at least in part on a difference between the first quantity of errors and the second quantity of errors and an operating voltage of the memory system associated with the first quantity of errors and the second quantity of errors satisfying a voltage threshold.

9. A memory system, comprising: one or more memory devices comprising one or more sets of memory cells; and one or more controllers coupled with the one or more memory devices and configured to cause the memory system to: determine a first quantity of errors for a first subset of a set of memory cells of the one or more sets of memory cells; determine a second quantity of errors for a second subset of the set of memory cells; and retire the set of memory cells based at least in

part on a difference between the first quantity of errors and the second quantity of errors satisfying a threshold and an operating temperature of the memory system associated with the first quantity of errors and the second quantity of errors satisfying a temperature threshold.
10. The memory system of claim 9, wherein: the set of memory cells is associated with a plurality of first access lines and a plurality of second access lines; the first subset of the set of memory cells is associated with a first subset of one or more of the plurality of first access lines; and the second subset of the set of memory cells is associated with a second subset of one or more of the plurality of first access lines.
11. The memory system of claim 9, wherein: to determine the first quantity of errors, the one or more controllers are configured to cause the memory system to determine a first quantity of uncorrectable errors associated with reading memory cells of the first subset of the set of memory cells; and to determine the second quantity of errors, the one or more controllers configured to cause the memory system to determine a second quantity of uncorrectable errors associated with reading memory cells of the second subset of the set of memory cells.
12. The memory system of claim 9, wherein the first quantity of errors, the second quantity of errors, or both are based at least in part on a quantity of flipped bits, a quantity of codewords that include errors, or a combination thereof.
13. The memory system of claim 9, wherein the one or more temperatures comprise an operating temperature of the one or more memory devices associated with the first quantity of errors and the second quantity of errors, wherein the one or more controllers are configured to cause the memory system to: retire the set of memory cells based at least in part on the operating temperature of the one or more memory devices satisfying a temperature threshold.
14. The memory system of claim 9, wherein the one or more controllers are configured to cause the memory system to: retire the set of memory cells based at least in part on an availability of a second set of memory cells of the one or more sets of memory cells; and replace the set of memory cells with the second set of memory cells based at least in part on retiring the set of memory cells.
15. The memory system of claim 9, wherein the one or more controllers are configured to cause the memory system to: retire the set of memory cells based at least in part on a difference between a write temperature and a read temperature associated with the first quantity of errors and the second quantity of errors satisfying a threshold temperature difference.
16. A memory system comprising: one or more memory devices comprising one or more sets of memory cells; and one or more controllers configured to cause the memory system to: determine a first quantity of errors for a first subset of a set of memory cells of the one or more sets of memory cells; determine a second quantity of errors for a second subset of the set of memory cells; and retire the set of memory cells based at least in part on a difference between the first quantity of errors and the second quantity of errors satisfying a threshold and an operating voltage of the one or more memory devices associated with the first quantity of errors and the second quantity of errors satisfying a voltage threshold.
17. A non-transitory computer-readable medium storing code comprising instructions which, when executed by one or more processors of an electronic device, cause the electronic device to: determine a first quantity of errors for a first subset of memory cells of a set of memory cells of the electronic device; determine a second quantity of errors for a second subset of memory cells of the set of memory cells; and retire the set of memory cells based at least in part on a difference between the first quantity of errors and the second quantity of errors satisfying a threshold and based at least in part on an operating temperature of the electronic device associated with the first quantity of errors and the second quantity of errors satisfying a temperature threshold.
18. The non-transitory computer-readable medium of claim 17, wherein: the instructions to determine the first quantity of errors, when executed by the one or more processors of the electronic device, cause the electronic device to determine the first quantity of errors associated with a first subset of access lines of a plurality of access lines of the set of memory cells; and the

instructions to determine the second quantity of errors, when executed by the one or more processors of the electronic device, cause the electronic device to determine the second quantity of errors associated with a second subset of access lines of the plurality of access lines of the set of memory cells.

19. The non-transitory computer-readable medium of claim 17, wherein the instructions, when executed by the one or more processors of the electronic device, cause the electronic device to: initiate a read scan of the set of memory cells based at least in part on an identified error condition of the set of memory cells; and determine the first quantity of errors and the second quantity of errors based at least in part on the read scan of the set of memory cells.